



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018 – 0380



Agricultural Machinery – Power-Operated Corn Sheller ACT MINI – KAIAO

Test Report Number	2018 – 0380
Date of Issuance	July 1, 2019
Valid Until	July 1, 2024
Agricultural Machinery	Power-Operated Corn Sheller
Type	Axial-flow with sifter and fan, open-cylinder , husker/sheller
Brand	ACT
Model	MINI – KAIAO
Test Requested by	ACT MACHINERIES & METALCRAFT CORPORATION Brgy. Turayong, Cauayan City, Isabela

SUMMARY

Performance Criteria	Requirement as per PAES 208:2000	AMTEC Test
Shelling recovery, %, minimum	97.0	99.43
Shelling efficiency, %, minimum	99.5	100
Losses, %, maximum		
a) Blower loss	1.0	0.21
b) Separation loss	1.0	0.36
c) Unshelled loss	0.5	0
d) Scattering loss	0.5	0
Purity, %, minimum	98	98.57
Mechanically damaged kernel, %, maximum	3.0	2.73
Net cracked kernel, %, maximum	5.0	2.83
Noise level, dB(A), maximum	100*	99.9

Note: *Allowable noise level for six (6) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983.

OBSERVATIONS

1. Operator's Manual	None
2. Previous valid test reports of similar brand and model	None
3. Prime mover used	7.46 kW SUPER KAIAO KA188FA Diesel Engine
4. Manufacturer	ACT MACHINERIES & METALCRAFT CORPORATION
5. Serial number	2263-18
6. Transporting of machine	Provided with transport wheels, hitch point and towing bars.
7. Adjustment of parts	Engine – cylinder belt tension can be adjusted. Air flow can be adjusted.
8. Safety features	No cover was provided in the V-belt-pulley transmission system.
9. Cleaning of concave and cylinder	With cover hinged to the body cleaning is possible.
10. Cleaning of fan component	Fan casing is fixed which make cleaning difficult.
11. Failure or abnormalities that observed on the sheller or its components during and after shelling operation	None
12. Number of operators	4 (one for hauling of test materials to the sheller, one for feeding of test materials, one for collecting and bagging of shelled grains, and one for stirring of mixture of grains and cobs on the oscillating screen)
13. Other remarks	No breakdown occurred.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	October 4, 2018
Location of Test	Turayong, Cauayan City, Isabela
Items Tested	Shelling capacity, Shelling recovery, Shelling efficiency, Speed of rotating components, Quality of work, Noise level, Fuel consumption, and Number of operators

METHODS OF TEST

The corn sheller was tested based on the Philippine National Standards PNS/PAES 209:2000 Agricultural Machinery – Power-Operated Corn Sheller – Methods of Test.

DESCRIPTION OF THE MACHINE



Figure 1. Parts of ACT MINI – KAIAO Corn Sheller powered by 7.46 kW SUPER KAIAO KA188FA Diesel Engine:
(a) Prime mover, (b) Belt-pulley transmission, (c) Shelling chamber
(d) Feeding table/ tray, (e) Blower, and (f) Transport wheel.

SPECIFICATIONS

Item		Manufacturer's Specification*	AMTEC Verification
1	Dimension and weight of sheller		
1.2	Overall length, mm	ND	1485
1.3	Overall width, mm	ND	1110
1.4	Overall height, mm	ND	1400
1.5	Weight of machine without engine, kg	ND	NM
2	Crop(s) for which machine is suitable	ND	Corn
3	Rated output capacity, kg/h	ND	1638
4	Recommended cylinder speed, rpm	ND	973
5	Engine		
5.1	Brand	ND	SUPER KAIAO
5.2	Model	ND	KA188FA
5.3	Serial number	ND	No Serial Number
5.4	Make	ND	ND
5.5	Rated power, kW	ND	7.46
5.6	Rated speed, rpm	ND	3600
5.7	Type	ND	Four-stroke, vertical single cylinder diesel engine
5.8	Weight	ND	ND
5.9	Starting system	ND	Rope starter
5.10	Cooling system	ND	Air-cooled
6	Type of power transmission system		
6.1	Engine to Shelling cylinder	V-Belt & Pulley	
6.1.1	Engine pulley diameter, mm	ND	51 x 1B
6.1.2	Shelling cylinder pulley diameter, mm	ND	254 x 1B
6.2	Shelling cylinder to Oscillating screen	V-Belt & Pulley	
6.2.1	Oscillating screen pulley diameter, mm	ND	179 x 1B
6.3	Oscillating screen to Fan	V-Belt & Pulley	
6.3.1	Fan pulley diameter, mm	ND	76 x 1B
6.4	Fan to auger	V-Belt & Pulley	
6.4.1	Auger pulley diameter, mm	ND	179 x 1B

Note: *The requesting party did not supply the complete machine specifications
 ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification*	AMTEC Verification
7	Shelling Cylinder		
7.1	Type	ND	Open
7.2	Size, L x D, mm	ND	510 x 360
7.3	Cylinder Teeth		
7.3.1	Type	ND	Knife – bar
7.3.2	Size, L x W x T, mm	ND	70 x 25 x 8
7.3.3	Number/Row	ND	5, 6, 5
7.3.4	Arrangement	ND	Alternating
7.3.5	Means of Attachment	ND	Welded to the cylinder
7.3.6	Material	ND	Mild steel
7.4	Cylinder cover material	ND	G.I. sheet #16
8	Fan		
8.1	Type	ND	Radial centrifugal
8.2	Number of units	ND	1
8.3	Fan for cleaning		
8.3.1	Length, mm	ND	540
8.3.2	Number of blades	ND	4
8.4	Material	ND	G.I. sheet #18
9	Oscillating Screen		
9.1	Dimension, L x W, mm	ND	635 x 540
9.2	Size of Perforations, mm	ND	12
9.3	Length of Stroke, mm	ND	15
9.4	Material	ND	Perforated G.I. metal sheet
10	Feeding table		
10.1	Shape	ND	Rectangular
10.2	Dimensions, L x W, mm	ND	665 x 410
10.3	Height from the ground, mm	ND	985
10.4	Mode of attachment	ND	Bolted
10.5	Material	ND	G.I. sheet #18

Note: *The requesting party did not supply the complete machine specifications

ND – No Data

SPECIFICATIONS (Cont'n)

Item	Manufacturer's Specification*	AMTEC Verification
11 Concave Component		
11.1 Overall Diameter, mm	ND	NM
11.2 Clearance, minimum, mm	ND	75
11.3 Material	ND	10 mm plain round bar
12 Transport Device		
12.1 Type	ND	Tires
12.2 Size	ND	ND
13 Safety Device	ND	None
14 Discharge Device	ND	Auger
15 Labor Requirement	ND	4
16 Adjustment	ND	Engine cylinder belt tension can be adjusted. Air flow can be adjusted.
17 Other special features	ND	Provide with hitched point and towing bars

Note: *The requesting party did not supply the complete machine specifications
 ND – No Data, NM – Not Measured

RESULTS

Item		Data	
Crop Condition			
1	Kind/Variety	ND	
2	Days after harvest	0	
3	Moisture content, %wb	23.13	
4	Size of corn – ear , L x D, mm	251 x 55	
5	Kernel-corn – ear ratio	0.79	
Performance Test			
1	Speed of components, rpm	Without load	With load
1.1	Prime mover	3548	3375
1.2	Shelling cylinder shaft	1057	746
1.3	Shelling cylinder peripheral speed, m/min	1011	714
1.4	Aspirator shaft	1756	1421
1.5	Oscillating screen shaft	762	520
1.6	Auger	769	537
2	Fan air velocity, suction, m/s	2.95	
3	Noise level, dB(A)		
3.1	Feeding operator	97.2	97.6
3.2	Bagger	98.3	99.9
4	Shelling time, min	6.7	
5	Shelled kernel, kg	239	
6	Actual shelling capacity, kg/h	2140.3	
7	Corrected shelling capacity, kg/h	2344.4	
8	Fuel time, min	15.5	
9	Fuel consumed, mL	343	
10	Fuel consumption rate, L/h	1.33	

RESULTS (Cont'n)**LABORATORY ANALYSIS**

	Item	Data
1	Shelling recovery, %	99.43
2	Shelling efficiency, %	100
3	Blower loss, %	0.21
4	Unshelled loss, %	0
5	Separation loss, %	0.36
6	Scattering loss, %	0
7	Purity of shelled kernel, %	98.57
8	Net cracked kernel, %	2.83
8.1	Manually shelled, %	0
8.2	Machine shelled	2.83
9	Mechanically damaged kernel, %	2.73

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